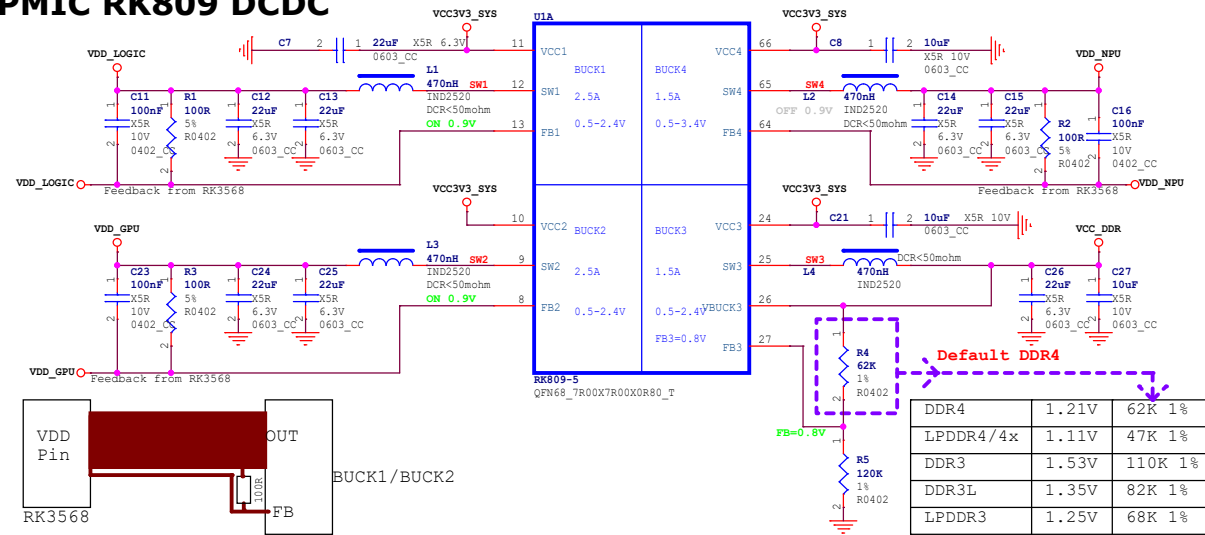


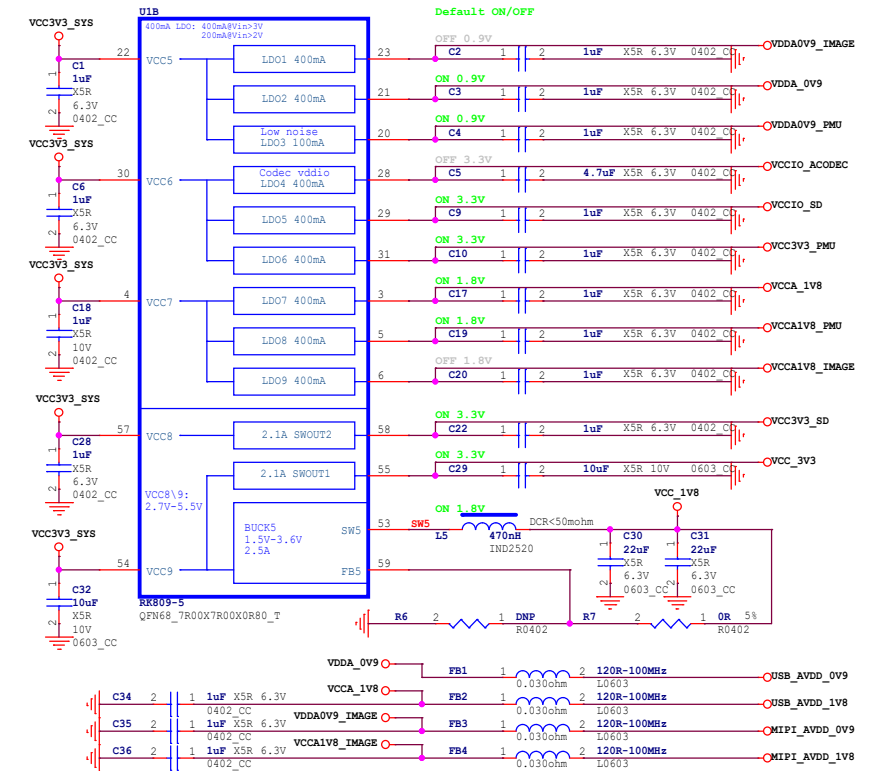
Revision History

Version	Date	By	Change Dscription	Approved
V1.0	2020-11-04	TOPEET	1:Revision preliminary version	
V1.1	2021-02-04	TOPEET	优化电源设计，引出所有GPIO	
V1.2	2023-11-01	TOPEET	更新LVDS1功能，修改连接器引脚定义	

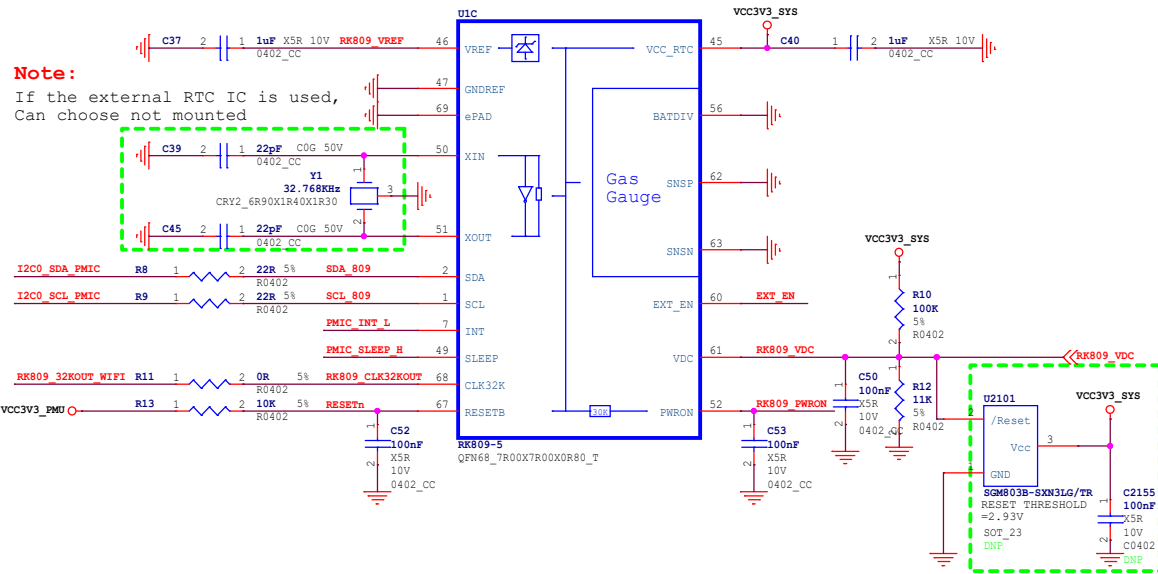
PMIC RK809 DCDC



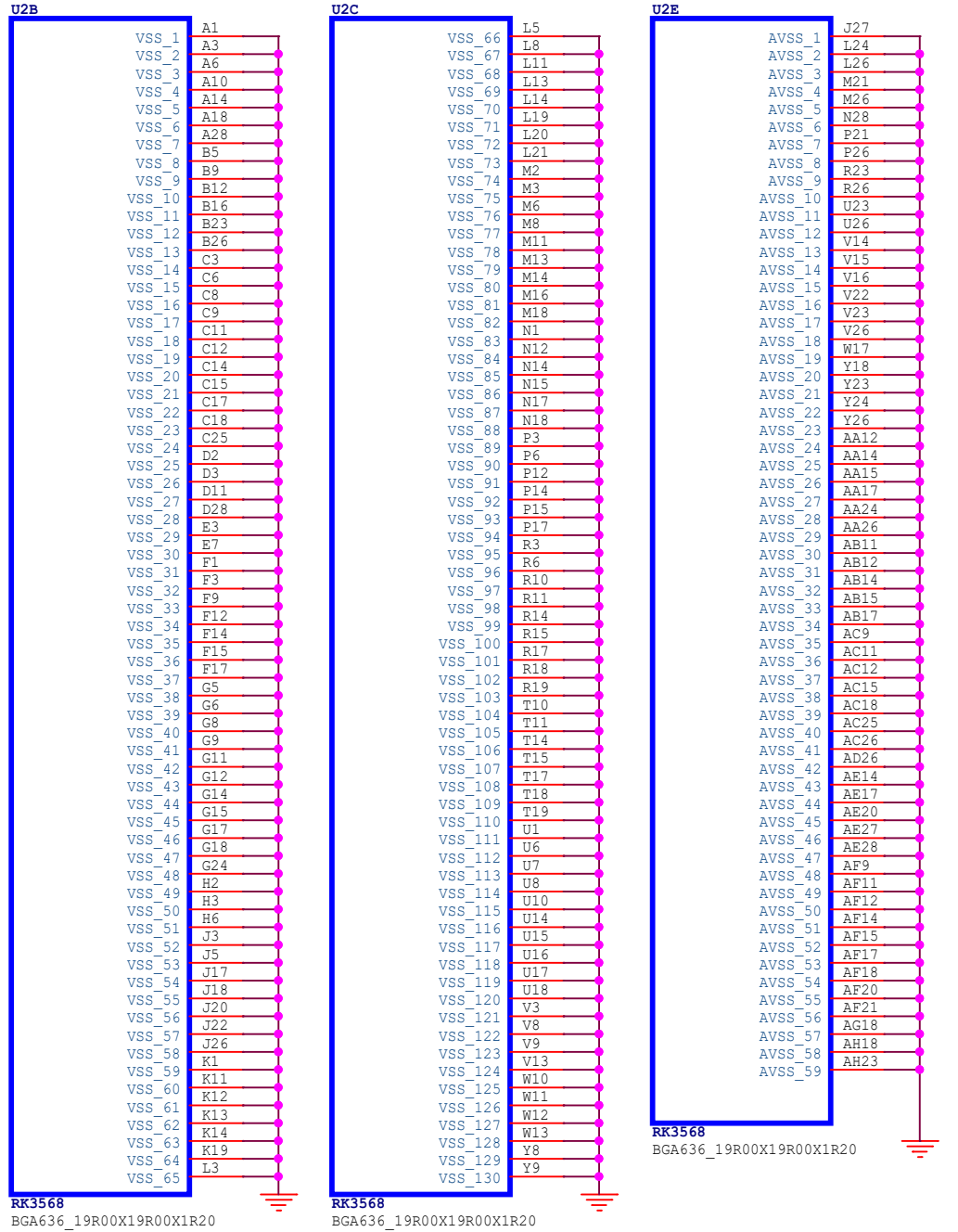
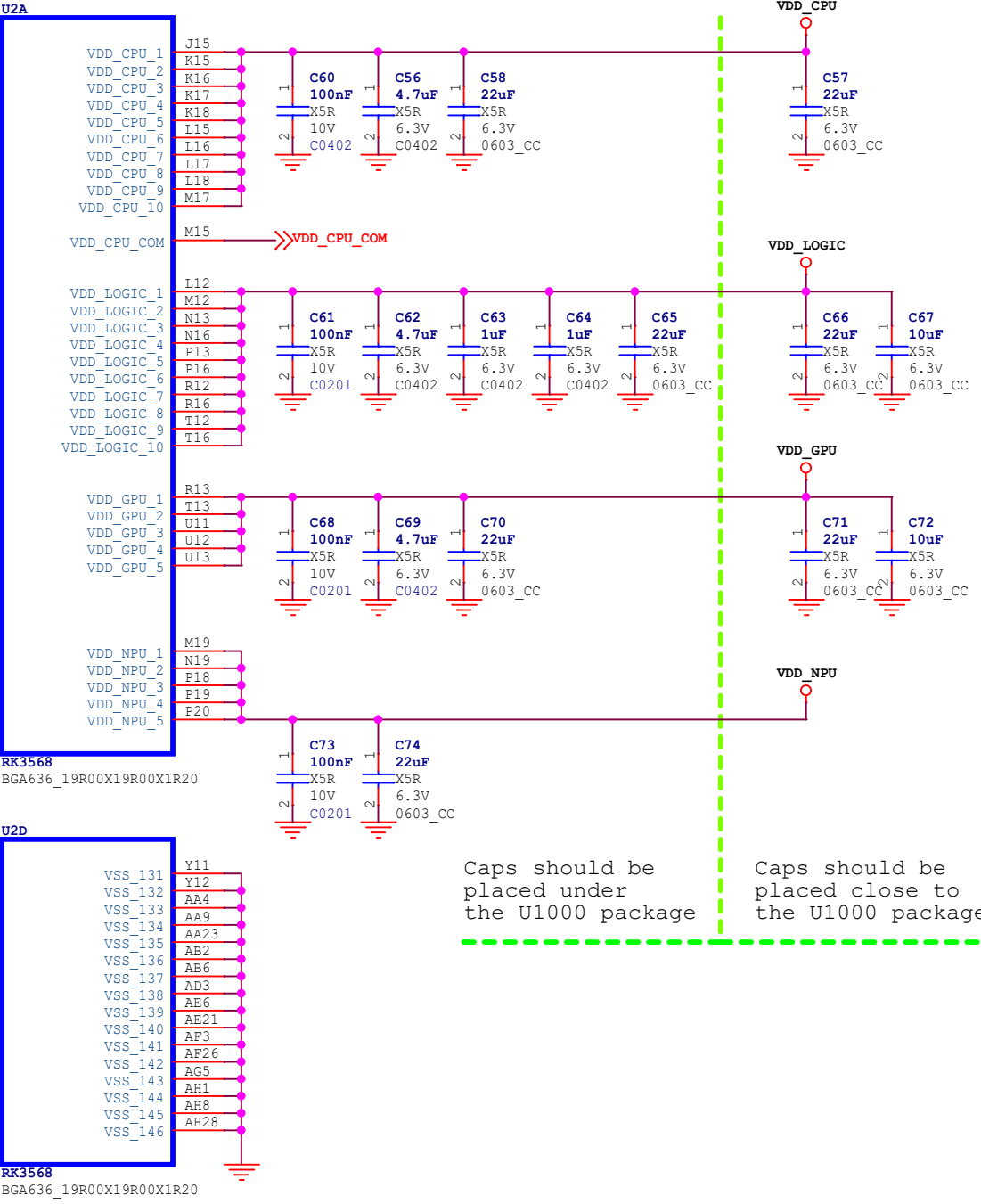
PMIC RK809 LDO



PMIC RK809 Management



RK3568_ABCDE (Power&Gnd)



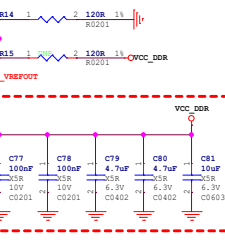
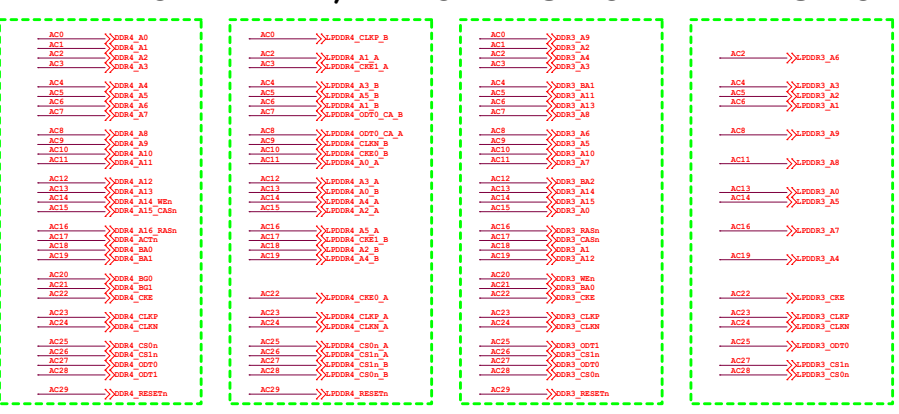
RK3568_F (DDR PHY)

DDR	DDR4	LPDDR4	DDR3	LPDDR3
D00	D00	D00	D00	D00
D01	D01	D01	D01	D01
D02	D02	D02	D02	D02
D03	D03	D03	D03	D03
D04	D04	D04	D04	D04
D05	D05	D05	D05	D05
D06	D06	D06	D06	D06
D07	D07	D07	D07	D07
D08	D08	D08	D08	D08
D09	D09	D09	D09	D09
D10	D10	D10	D10	D10
D11	D11	D11	D11	D11
D12	D12	D12	D12	D12
D13	D13	D13	D13	D13
D14	D14	D14	D14	D14
D15	D15	D15	D15	D15
D16	D16	D16	D16	D16
D17	D17	D17	D17	D17
D18	D18	D18	D18	D18
D19	D19	D19	D19	D19
D20	D20	D20	D20	D20
D21	D21	D21	D21	D21
D22	D22	D22	D22	D22
D23	D23	D23	D23	D23
D24	D24	D24	D24	D24
D25	D25	D25	D25	D25
D26	D26	D26	D26	D26
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D35	D35	D35	D35	D35
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D96	D96	D96	D96	D96
D97	D97	D97	D97	D97
D98	D98	D98	D98	D98
D99	D99	D99	D99	D99

DDR4	LPDDR4	DDR3	LPDDR3
D00	D00	D00	D00
D01	D01	D01	D01
D02	D02	D02	D02
D03	D03	D03	D03
D04	D04	D04	D04
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D98	D98	D98	D98
D99	D99	D99	D99

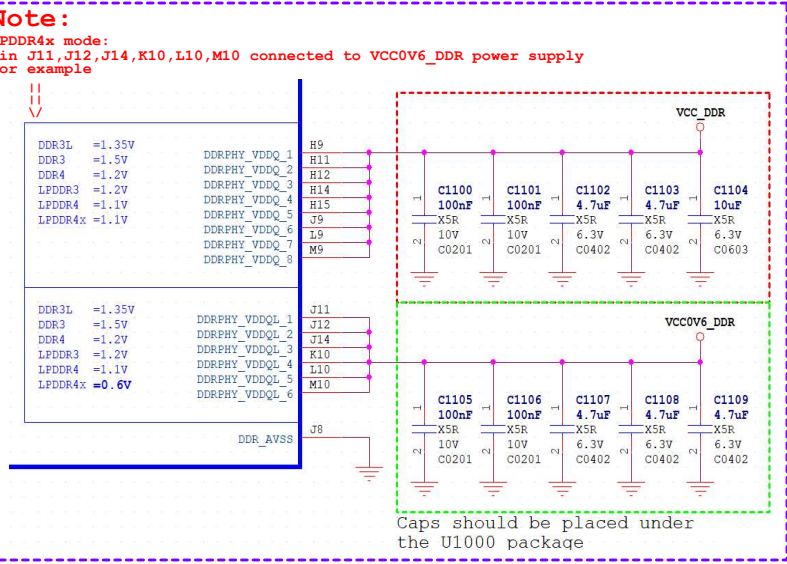
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D01	D01	D01	D01
D02	D02	D02	D02
D03	D03	D03	D03
D04	D04	D04	D04
D05	D05	D05	D05
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D09	D09	D09	D09
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D36	D36	D36	D36
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D95	D95	D95	D95
D96	D96	D96	D96
D97	D97	D97	D97
D98	D98	D98	D98
D99	D99	D99	D99

Default DDR4 AC LPDDR4/4x AC DDR3 AC LPDDR3 AC

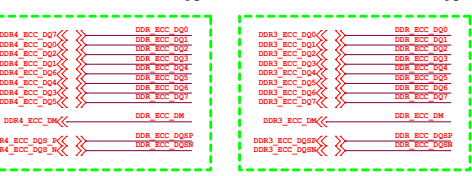


Default DDR4

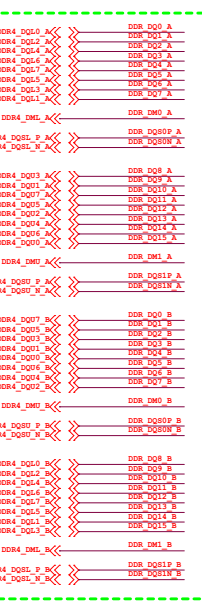
Note:
Caps should be placed under the U1000 package



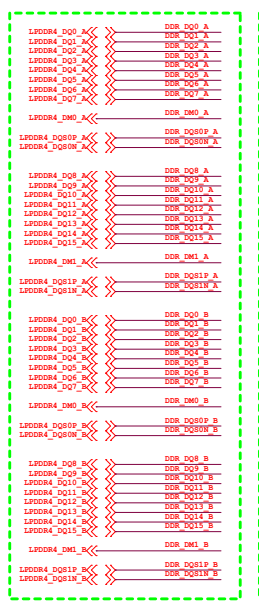
DDR4 ECC DQ DDR3 ECC DQ



Default DDR4 DQ



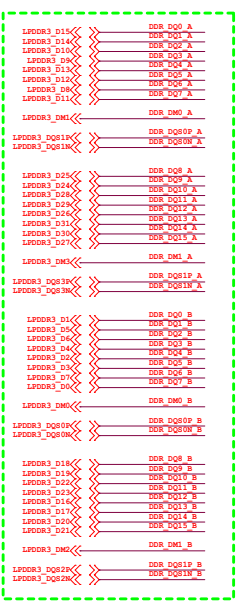
LPDDR4/4x DQ



DDR3 DQ



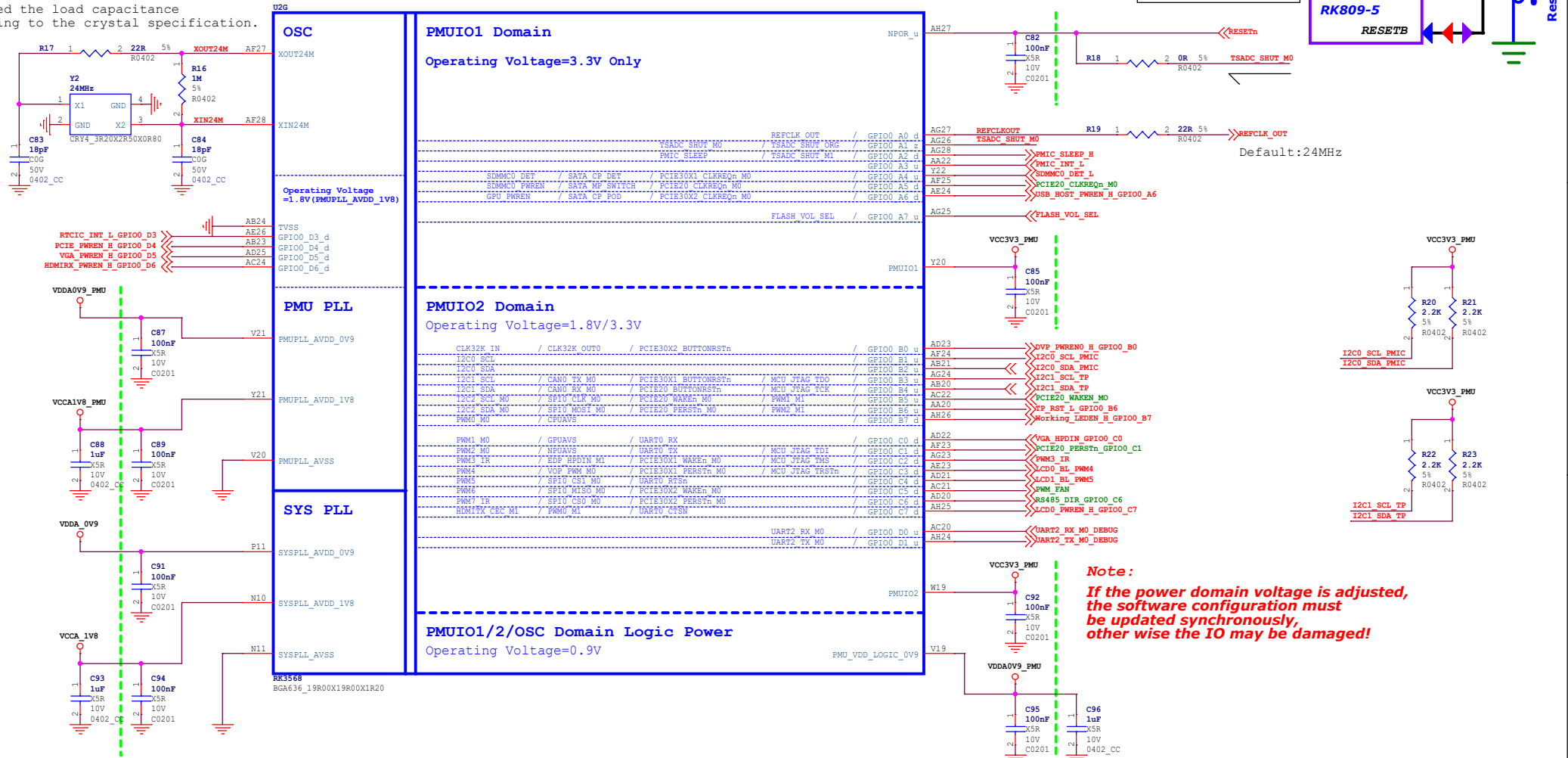
LPDDR3 DQ



RK3568_G (OSC/PLL/PMUIO1/2)

Note:

Adjusted the load capacitance according to the crystal specification.

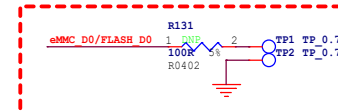
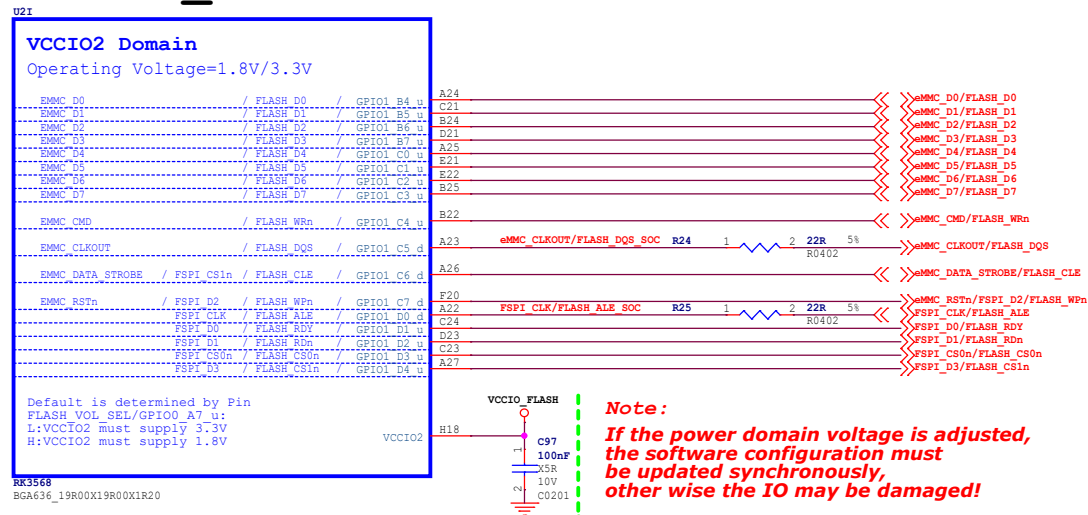


Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

TOPEET 讯为 北京迅为电子有限公司				
Project:	iTOP-3568			
File:	topeet_rk3568_coreaddr2_v1_1			
Date:	Friday, November 20, 2020	Rev:	V1.2	
Designed by:	TOPEET	Reviewed by:	TOPEET	Sheet: 6 of 20

RK3568_I (VCCIO2 Domain)

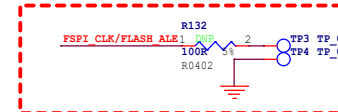


Note:

For eMMC or Nand Flash:
If eMMC_D0/FLASH_D0=0V at after power on and reset, then system will enter into Maskrom mode.

Layout note:

Test point must be placed on the line, and no branch can be added



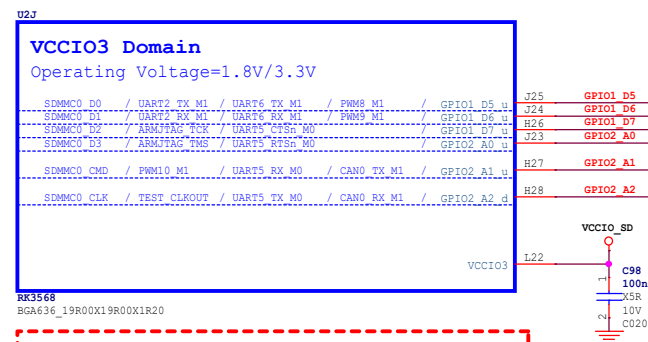
Note:

For SPI Flash:
If FSPI_CLK=0V at after power on and reset, then system will enter into Maskrom mode.

Note:
Reserve TestPoint for put the system into Maskrom mode to update the firmware
When writing mismatched firmware or other conditions result in boot failure, use this test point

Except in this case, please use Recovery Key
Put the system into loader mode to update the firmware

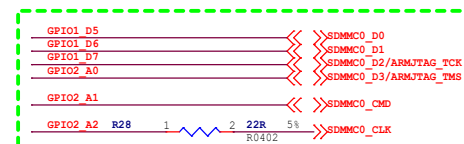
RK3568_J (VCCIO3 Domain)



Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

Default SDMMC0 & JTAG

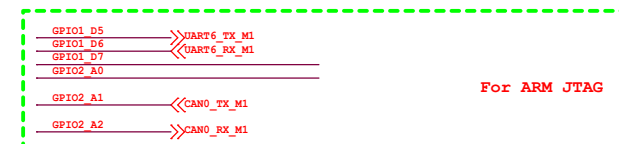


Note:

If the power domain voltage is adjusted, the software configuration must be updated synchronously, other wise the IO may be damaged!

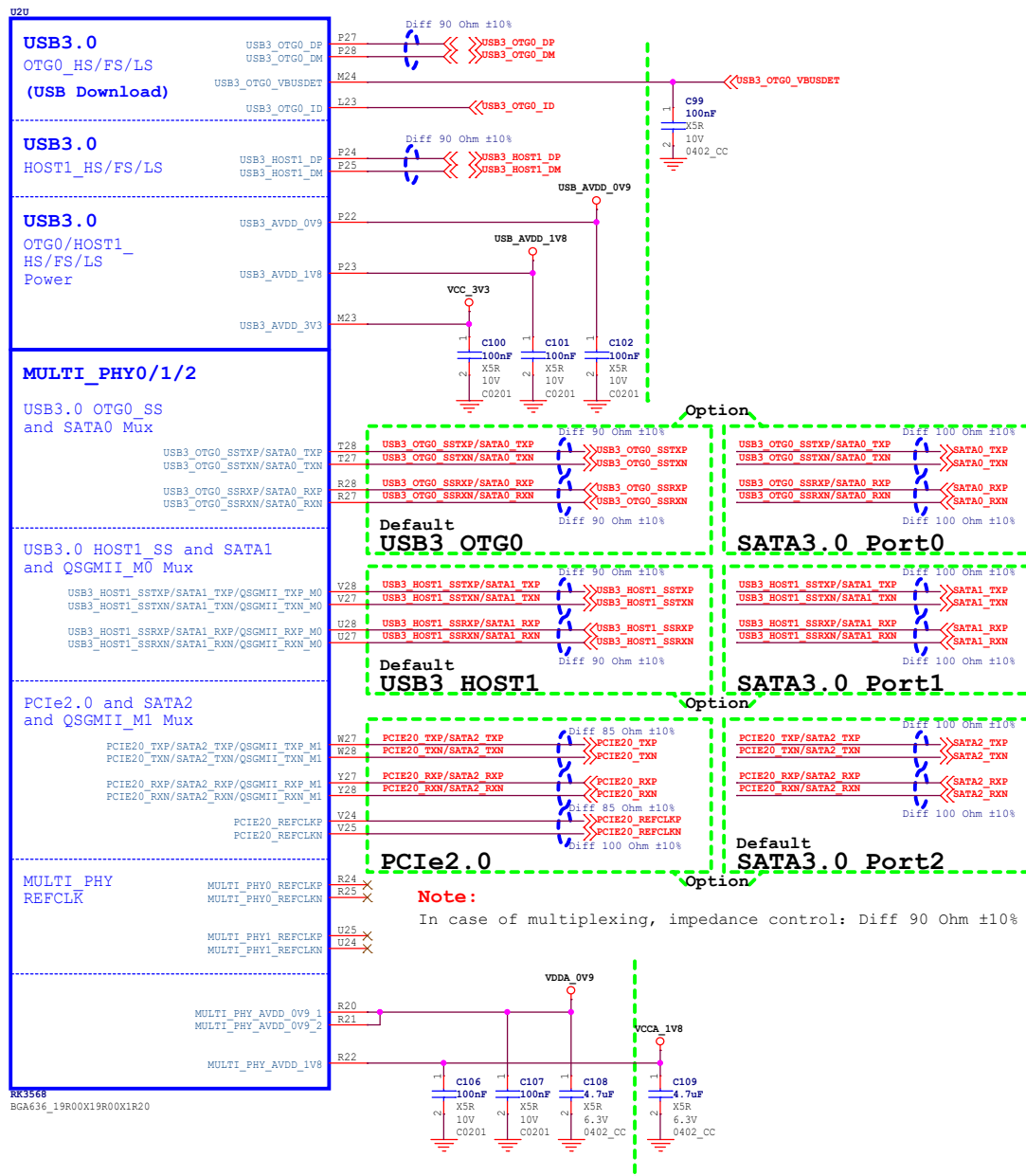
Option

If no MicroSD Card function: UART&CAN&JTAG



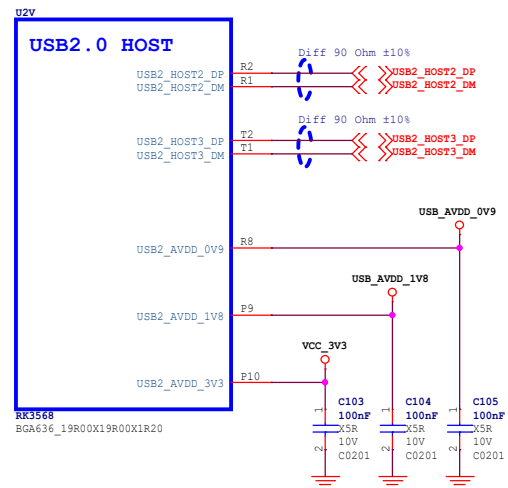
For ARM JTAG

RK3568_U (USB3.0/SATA/QSGMII/PCIE2.0 x1)

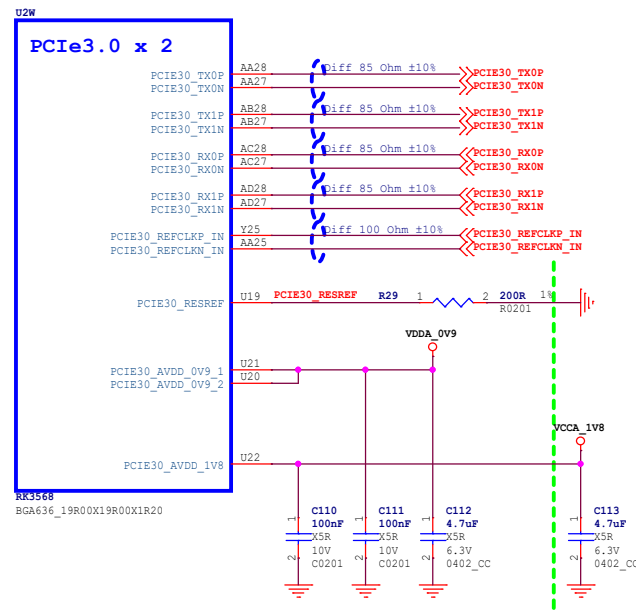


Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3568_V (USB2.0 HOST)



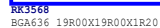
RK3568_W (PCIE3.0 x2)



TOPEET 讯为 北京迅为电子有限公司

Project:	ITOP-3568
File:	topeet_rk3568_coreddr4x2_v1_1
Date:	Friday, November 20, 2020
Designed by:	TOPEET
Reviewed by:	TOPEET
Rev:	V1.2
Sheet:	8 of 20

U2K



```

GPIO2_A3 _____>>>SDMMC1_D0
GPIO2_A4 _____>>>SDMMC1_D1
GPIO2_A5 _____>>>SDMMC1_D2
GPIO2_A6 _____>>>SDMMC1_D3

GPIO2_A7 _____>>>SDMMC1_CMD

GPIO2_B0 _____>>>SDMMC1_CLK

GPIO2_B1 _____>>>I2C4_SDA_M1
GPIO2_B2 _____>>>I2C4_SCL_M1

GPIO2_B3 _____>>>UART1_RX_M0
GPIO2_B4 _____>>>UART1_TX_M0
GPIO2_B5 _____>>>UART1_RTSn_M0
GPIO2_B6 _____>>>UART1_CTSn_M0

GPIO2_B7 _____>>>BT_REG_ON_H_GPIO2_B7
GPIO2_C0 _____>>>BT_WAKE_HOST_H_GPIO2
GPIO2_C1 _____>>>HOST_WAKE_BT_H_GPIO2

GPIO2_C2 _____>>>SOC_PCM_CLK
GPIO2_C3 _____>>>SOC_PCM_SYNC
GPIO2_C4 _____>>>SOC_PCM_OUT
GPIO2_C5 _____>>>SOC_PCM_IN

GPIO2_C6 _____>>>CLK32K_UART1_WIFI

```

Note:

Option

GPIO2_A3 **2 22R 5% R0402** **GMAC0_RXMD2**
GPIO2_A4 **2 22R 5% R0402** **GMAC0_RXD3**
GPIO2_A5 **2 22R 5% R0402** **GMAC0_RXCLK**
GPIO2_A6 **R30** **1** **2 22R 5% R0402** **GMAC0_TXD2**
GPIO2_A7 **R31** **1** **2 22R 5% R0402** **GMAC0_TXD3**
GPIO2_B0 **R33** **1** **2 22R 5% R0402** **GMAC0_TXCLK**

GPIO2_B3 **R34** **1** **2 22R 5% R0402** **GMAC0_TXD0**
GPIO2_B4 **R35** **1** **2 22R 5% R0402** **GMAC0_TXD1**
GPIO2_B5 **R36** **1** **2 22R 5% R0402** **GMAC0_TXD2**
GPIO2_B6 **1** **2 22R 5% R0402** **GMAC0_TXEN**

GPIO2_B7 **1** **2 22R 5% R0402** **GMAC0_RXD0**
GPIO2_C0 **1** **2 22R 5% R0402** **GMAC0_RXDV_CRS**
GPIO2_C1 **R37** **1** **2 22R 5% R0402** **ETH0_RXCLK_25**

GPIO2_C2 **1** **2 22R 5% R0402** **GMAC0_MCLKINOUT**
GPIO2_C3 **1** **2 22R 5% R0402** **GMAC0_MDC**
GPIO2_C4 **1** **2 22R 5% R0402** **GMAC0_MDIO**
GPIO2_C5 **GMAC0_RXER** **1** **2 22R 5% R0402** **GMAC0_RXER**

VCCIO4 **R39** **1** **2 0R 5%** **VCC_1V8**
R41 **1** **2 0R 5%** **VCC_3V3**

Note:

If Ethernet PHY uses other models, please note whether the default pull-up and pull-down of GPIO affect Ethernet PHY function

At present,
TXEN will be affected
if it defaults to high level
need to add a 4.7K
resistance to ground

Note:

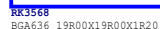
According to the actual choice of mounted

- Cannot be mounted at the same time

Default:1.8V

Select the voltage according to the application

U2N



Note:

When use HDMI, HDMITX_SCL/SDA cannot be shared with other devices

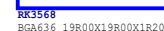
Note:

If the power domain voltage is adjusted, the software configuration must be updated synchronously, other wise the IO may be damaged!

GPIO4_C5 >>> UART9_TX_M1 GPIO4_C5 >>> SATA1_ACT_LED
GPIO4_C6 >>> UART9_RX_M1 GPIO4_C6 >>> SATA0_ACT_LED

Option

U2



Note:

Must be mounted

Note:

If there is no Key requirement,
two test points must be reserved
to facilitate firmware update

It is suggested to reserve a
Key to facilitate the
development debug

If SARADC_VIN0=0V
at after power on and reset,
then system will
enter into loader mode.

Note:

Caps of between dashed green lines
and U1000 should be placed under
the U1000 package

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Project:	ITOP-3568				
File:	topest_rk3568_coreaddr4x2_v1_1				
Date:	Friday, November 20, 2020			Rev:	V1.2
Designed by:	TOPEET	Reviewed by:	TOPEET	Sheet:	9 of 20

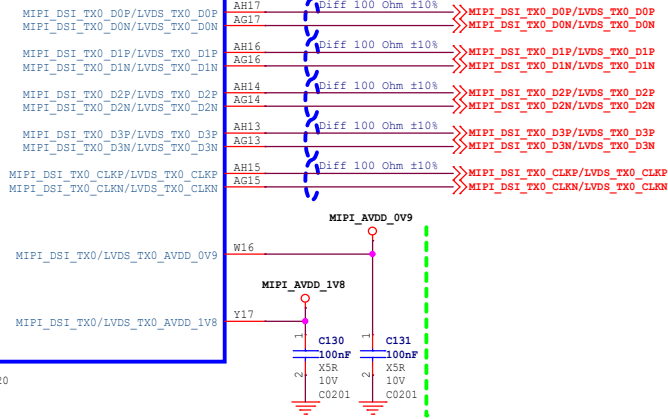
[illegible][illegible]

Note:
According to the actual choice of mounted
Cannot be mounted at the same time

RK3568_R (MIPI_DSI_TX0/LVDS_TX0)

U2R

MIPI DSI TX0/LVDS TX0

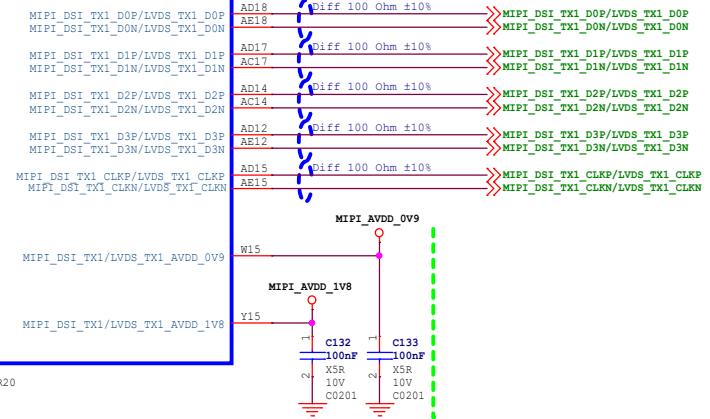


RK3568
BGA636_19R00X19R00X1R20

RK3568_S (MIPI_DSI_TX1/LVDS_TX1)

U2S

MIPI DSI TX1/LVDS TX1



RK3568
BGA636_19R00X19R00X1R20

Note:

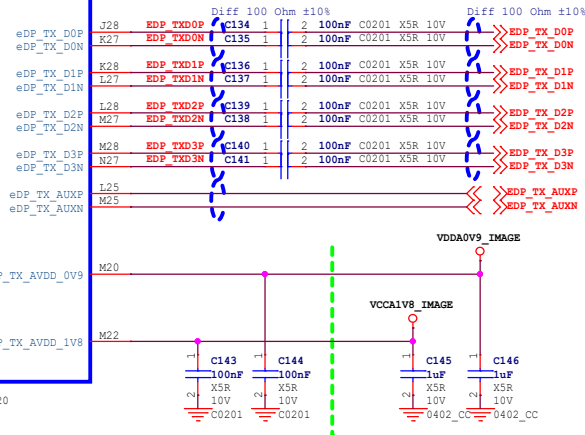
MIPI DSI TX0 + MIPI DSI TX1 support dual MIPI DSI TX interface, Left and Right support redistribution.

LVDS_TX0 + LVDS_TX1 support dual LVDS TX interface, Left and Right support redistribution

RK3568_T (eDP TX)

U2T

eDP_TX



RK3568
BGA636_19R00X19R00X1R20

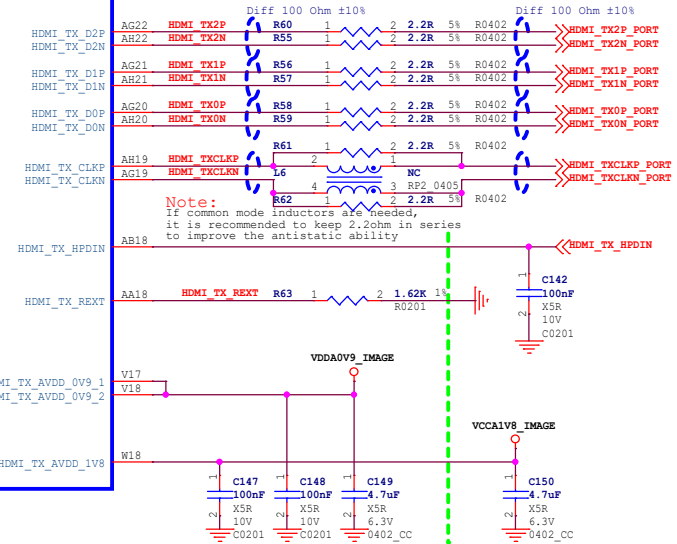
Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

RK3568_Q (HDMI2.0 TX)

U2Q

HDMI2.0 TX



RK3568
BGA636_19R00X19R00X1R20

Note:
If common mode inductors are needed, it is recommended to keep 2.2ohm in series to improve the antistatic ability

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Project:	ITOP-3568				
File:	topeet_rk3568_coreddr4x2_v1_1				
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D



B



1

RK3568_H (VCCIO1 Domain)

U2H

VCCIO1 Domain

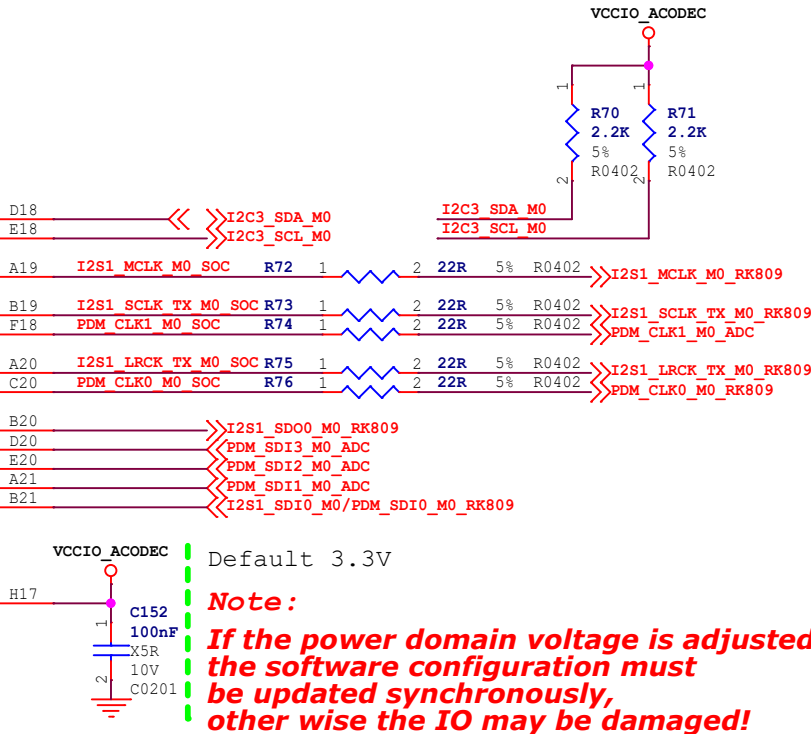
Operating Voltage=1.8V/3.3V

I2C3 SDA M0	/ UART3 RX M0	/ CAN1 RX M0	/ AUDIOPWM LOUT P / ACODEC ADC DATA	/ GPIO1 A0 u
I2C3 SCL M0	/ UART3 TX M0	/ CAN1 TX M0	/ AUDIOPWM LOUT N / ACODEC ADC CLK	/ GPIO1 A1 u
I2S1 MCLK M0	/ UART3 RTSn M0	/ SCR CLK	/ PCIE30X1 PERSTn M2	/ GPIO1 A2 d
I2S1 SCLK TX M0	/ UART3 CTSn M0	/ SCR IO	/ PCIE30X1 WAKEn M2	/ ACODEC DAC CLK
I2S1 SCLK RX M0	/ UART4 RX M0	/ PDM CLK1 M0	/ SPDIF TX M0	/ GPIO1 A4 d
I2S1 LRCK TX M0	/ UART4 RTSn M0	/ SCR RST	/ PCIE30X1 CLKREQn M2	/ ACODEC DAC SYNC
I2S1 LRCK RX M0	/ UART4 TX M0	/ PDM CLK0 M0	/ AUDIOPWM ROUT P	/ GPIO1 A6 d
I2S1 SDO0 M0	/ UART4 CTSn M0	/ SCR DET	/ AUDIOPWM ROUT N / ACODEC DAC DATAL	/ GPIO1 A7 d
I2S1 SDO1 M0	/ I2S1 SDI3 M0	/ PDM SDI3 M0	/ PCIE20 CLKREQn M2	/ ACODEC DAC DATAR
I2S1 SDO2 M0	/ I2S1 SDI2 M0	/ PDM SDI2 M0	/ PCIE20 WAKEn M2	/ ACODEC ADC SYNC
I2S1 SDO3 M0	/ I2S1 SDI1 M0	/ PDM SDI1 M0	/ PCIE20 PERSTn M2	/ GPIO1 B2 d
	/ I2S1 SDI0 M0	/ PDM SDI0 M0		/ GPIO1 B3 d

RK3568
BGA636_19R00X19R00X1R20

Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package



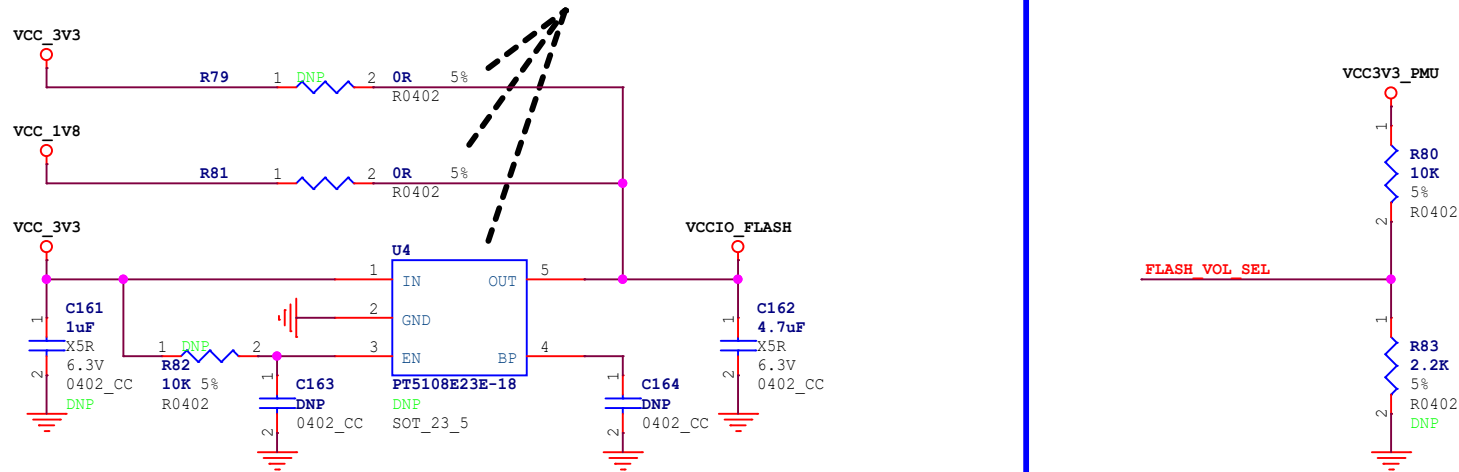
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Project:	ITOP-3568				
File:	topeet_rk3568_coreddr4x2_v1_1				
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Flash Power Manage

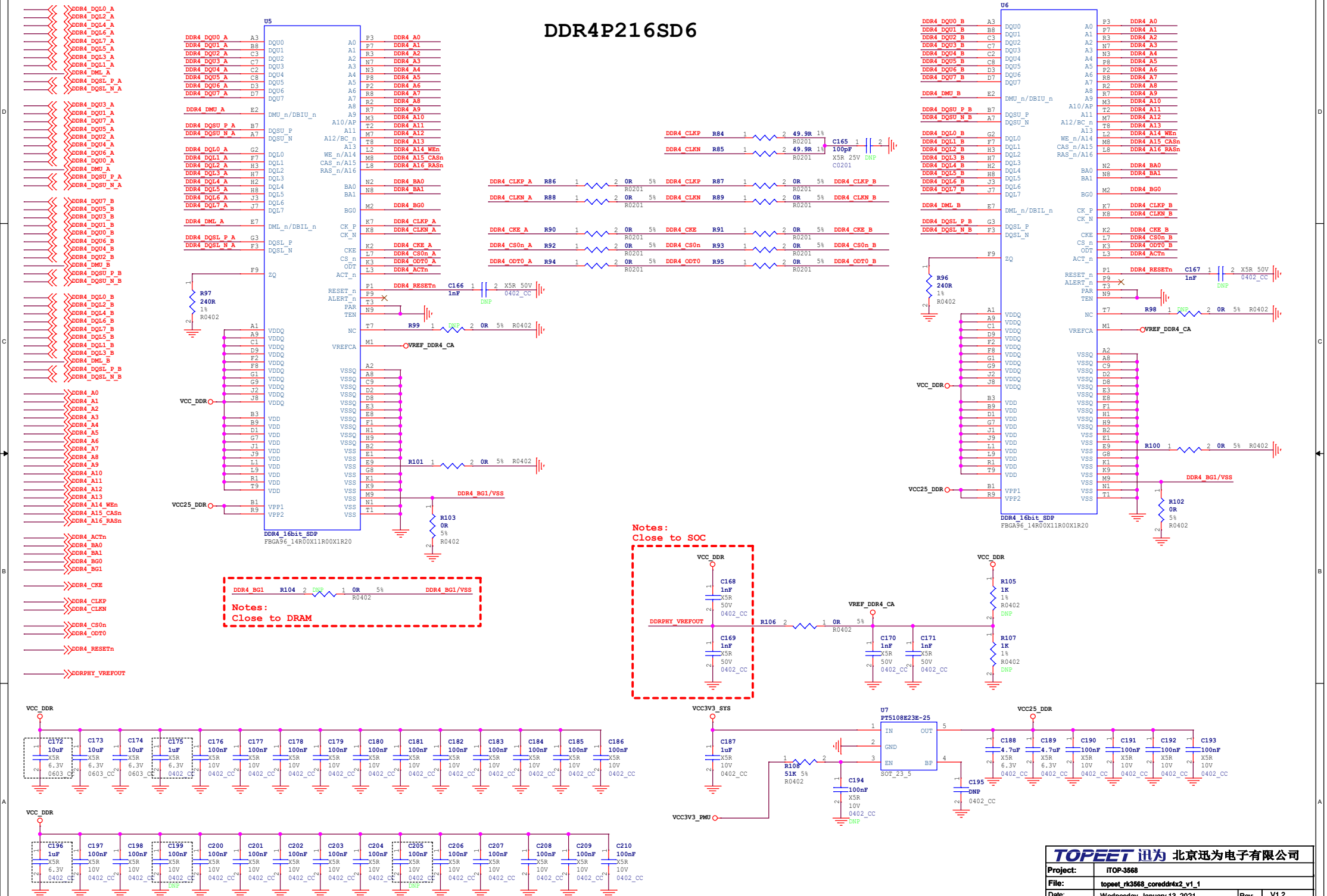
Note:

According to the actual choice of mounted
Cannot be mounted at the same time

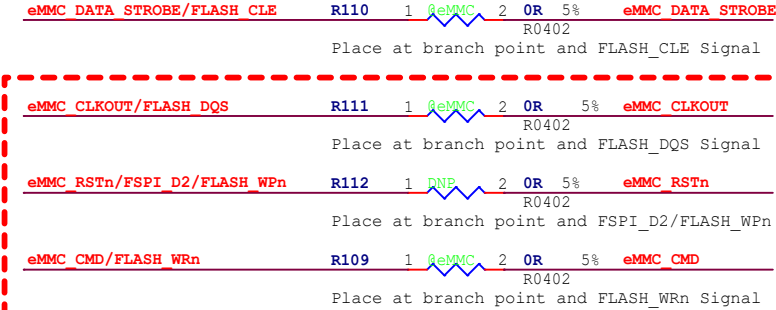
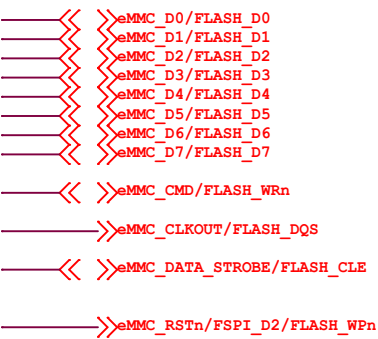


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Project:	ITOP-3568		
File:	topeet_rk3568_coreddr4x2_v1_1		
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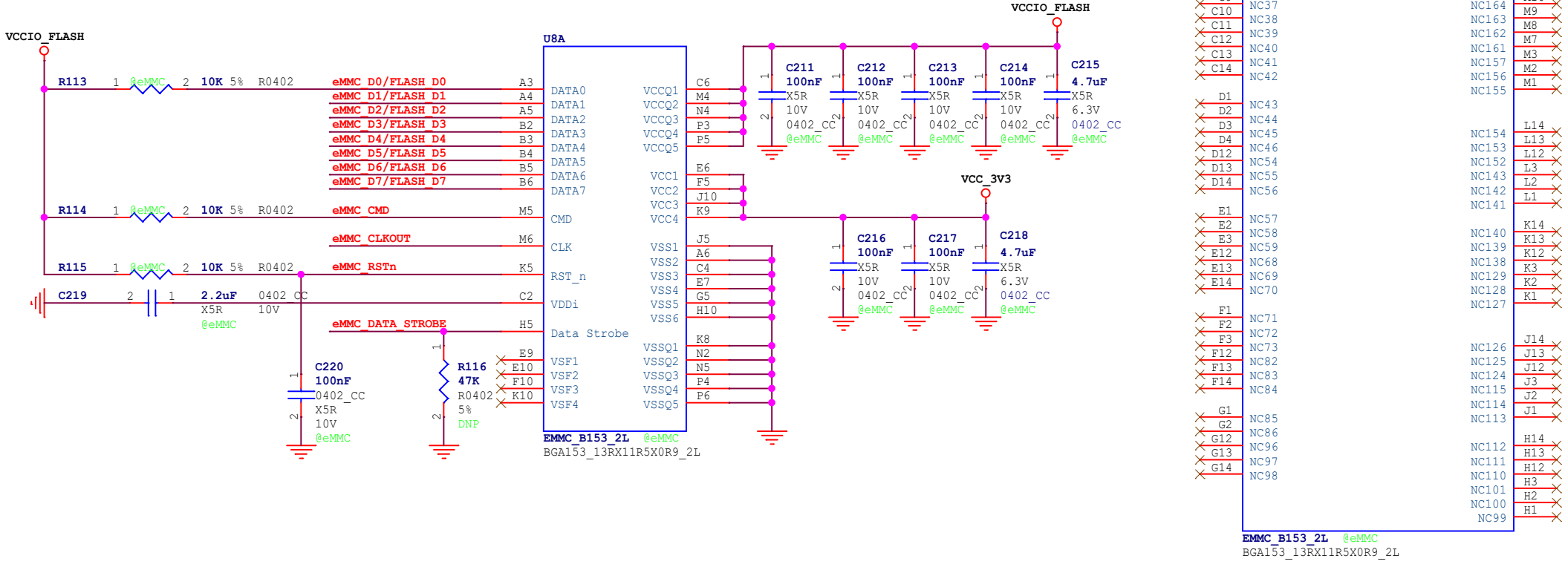
DDR4P216SD6



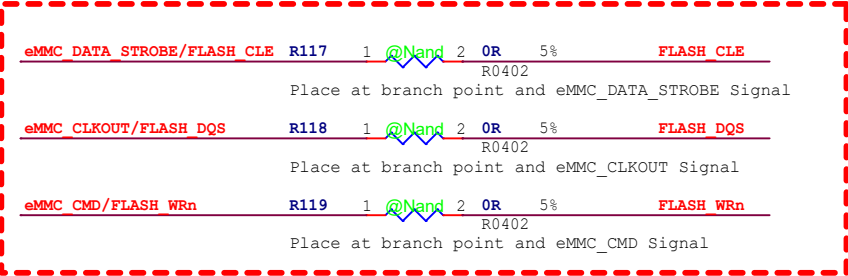
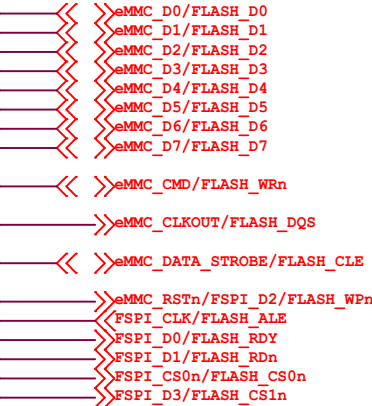
eMMC Flash



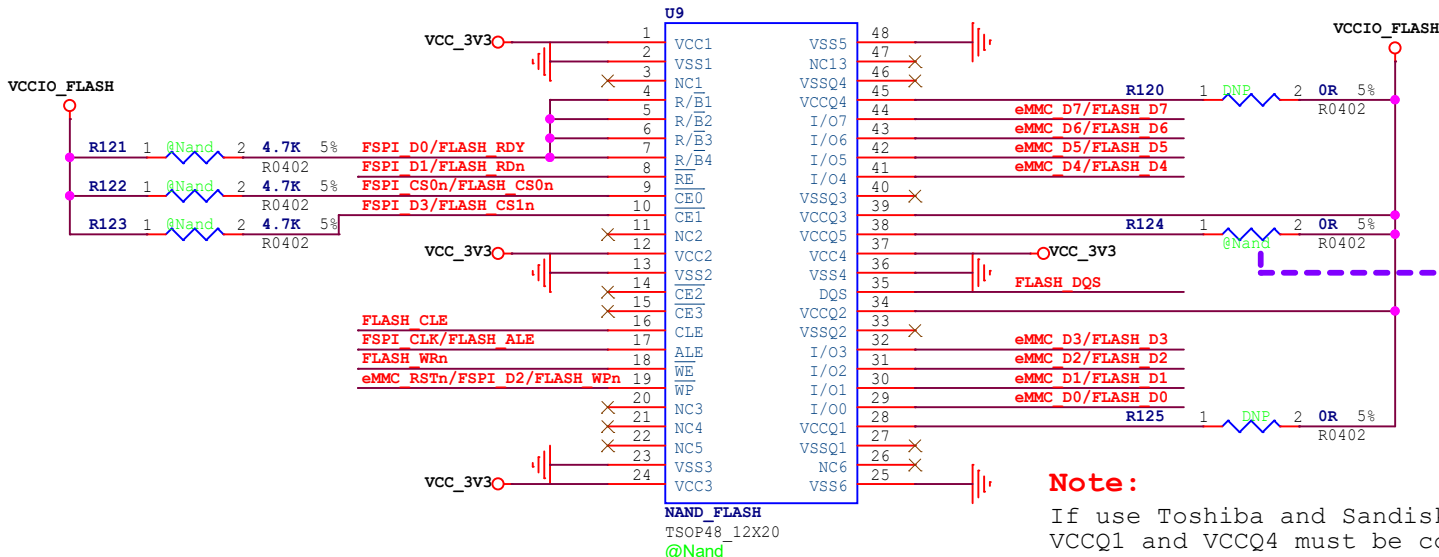
Note:
No need to double layout with Nand Flash, 0R resistor can be omitted



Nand Flash

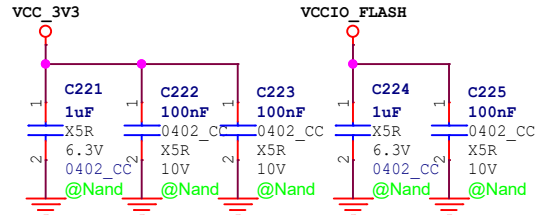


Note:
No need to double layout with eMMC, 0R resistor can be omitted

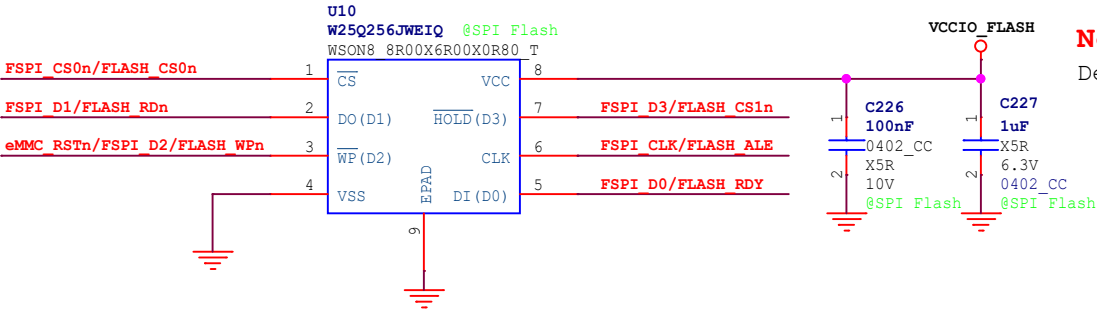
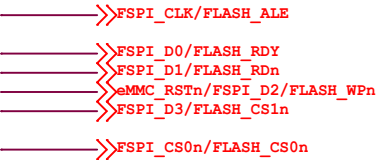


Note:
If use SLC Nand, This Resistance is DNP

Note:
If use Toshiba and Sandisk DDR mode, VCCQ1 and VCCQ4 must be connected to VCC_IO.

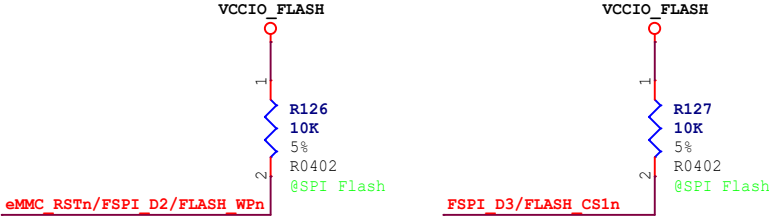


SPI Flash



Note:
Default: 1.8V

Support:
1bit SPI NOR or SPI NAND
4bit SPI NOR or SPI NAND



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File:	topeet_rk3568_coreddr4x2_v1_1			
Date:	Friday, November 20, 2020		Rev:	V1.2
Designed by:	TOPEET	Reviewed by:	TOPEET	Sheet: 19 of 20

